Approximate Bayesian Computation: A Practical Guide to Methods, Implementation, and Diagnostics

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1 Introduction

Bayesian inference requires evaluating the likelihood $f(y \mid \theta)$, which is analytically intractable for a wide range of situations, such as ecological population dynamics, or epidemic spread on networks. In these settings, the model can be simulated for any proposed parameter value, but the probability of observing a specific dataset cannot be computed in closed form.

Approximate Bayesian Computation (ABC) is a form of likelihood-free inference method which resolves this by replacing likelihood evaluation with simulation-based comparison. Rather than computing $f(y_{obs} \mid \theta)$ directly, ABC asks whether the data simulated for a given θ resembles the observed data. This report provides a self-contained treatment of ABC methods, progressing from the basic rejection algorithm through advanced sampling and post-processing techniques. All methods are implemented in a Python library and validated on a stochastic Lotka-Volterra predator-prey intractable model, with an MA(2) time series model serving as a tractable validation backbone throughout. The library is accessible at https://github.com/KevinQ989/abclib.

2 ABC Foundations

2.1 The Bayesian Framework and the Intractability Problem

Bayesian inference aims to update a prior belief $\pi_0(\theta)$ about parameters θ given some observed data y_{obs}, to get a posterior distribution $\pi(\theta)$ via Bayes' theorem:

$$\pi(\theta \mid y_{obs}) \propto \pi_0(\theta) \times f(y_{obs}|\theta).$$

However, for many models of practical interest, the likelihood $f(y_{obs} \mid \theta)$ cannot be evaluated. This occurs when the data-generating process involves latent variables, stochastic dynamics, or combinatorial structures that make the probability of any specific observation analytically intractable. Standard posterior sampling methods such as MCMC require likelihood evaluations at every step and therefore cannot be applied directly.

2.2 The ABC Framework

ABC replaces likelihood evaluation with a simulation-based comparison. For a given tolerance $\varepsilon > 0$ and distance function $d(\cdot, \cdot)$, the ABC posterior is defined as:

$$\pi_{ABC}(\theta \mid y_{obs}) \propto \pi_0(\theta) \times \mathbf{1}\left[d(S(y_{sim}), S(y_{obs})) \leq \varepsilon\right],$$

where $y_{sim} \sim f(\cdot \mid \theta)$ is data simulated under θ and $S(\cdot)$ is a summary function mapping data to a lower-dimensional vector of summary statistics. As $\varepsilon \to 0$ and $S(\cdot)$ is sufficient for θ, the ABC posterior converges to the true posterior. In practice, ε is kept small but non-zero for computational tractability, introducing a controlled approximation error.

The ABC simulation loop proceeds as follows:

1. Draw a candidate $\theta^* \sim \pi_0(\theta)$ from the prior.
2. Simulate data $y_{sim} \sim f(\cdot \mid \theta^*)$ from the model.
3. Compare summary statistics $s_{sim} = S(y_{sim})$ and $s_{obs} = S(y_{obs})$.
4. Accept the θ^* if $d(s_{sim}, s_{obs}) \leq \varepsilon$, otherwise reject.

The accepted draws form a sample from the ABC posterior.

2.3 Summary Statistics

Comparing y_{sim} to y_{obs} directly is impractical, especially in high dimensions, and sensitive to irrelevant variation. The summary function $S(\cdot)$ maps the full dataset to a lower-dimension summary vector, retaining only the information relevant to inferring θ. If $S(\cdot)$ is sufficient for θ and captures all information in the data about the parameters, then replacing the data with summaries introduces no additional approximation error beyond the tolerance. In practice, exact sufficiency is rarely achievable and summary statistics must be designed to be approximately sufficient. The design of summary statistics is discussed further in Section 3.

2.4 Distance Functions

The distance function $d(s_{obs}, s_{sim})$ quantifies how closely the simulated summaries match the observed. Two common choices are:

Euclidean distance.

$$d = \|s_{obs} - s_{sim}\|_2 = \sqrt{\sum_i (s_{obs,i} - s_{sim,i})^2}$$

Simple and computationally cheap, but treats all summary statistics as independent and equally scaled.

Mahalanobis distance.

$$d = \sqrt{(s_{obs} - s_{sim})^T \Sigma^{-1} (s_{obs} - s_{sim})}$$

where Σ is the covariance matrix of $S(Y)$ estimated from prior predictive simulations. Accounts for the scale and correlation structure of the summaries, but requires estimating Σ, which is unreliable in moderate to high dimensions.

2.5 Normalisation

When summary statistics operate on different scales, those with larger magnitudes will dominate the distance calculation, causing other statistics to be effectively ignored. To prevent this, each summary statistic s_i is normalised before distance is computed. A standard approach is to scale by the prior-predictive standard deviation:

$$\tilde{s}_i = \frac{s_i}{\mathrm{Std}(s_i)}$$

where the standard deviation is estimated from simulations drawn under the prior. This places all statistics on a comparable scale without requiring knowledge of the true posterior.

2.6 Tolerance Selection

The tolerance ε controls the trade-off between approximation accuracy and computational cost. A small ε yields a posterior close to the true posterior, but accepts very few draws, thus requiring a large simulation budget. A large ε accepts more draws but introduces greater approximation error.

In practice, ε is often chosen as a quantile of the empirical distance distribution, for example, accepting the closest $q = 5\%$ of simulated draws. This decouples the threshold from the absolute scale of the distance function and automatically adapts to the problem. The sensitivity of the posterior to the choice of q should always be checked.

2.7 Notation Summary

For reference throughout this report, Table 1 defines the core notation used.

Symbol	**Definition**
θ	Parameter vector
y_{obs}	Observed data
y_{sim}	Simulated data under θ
$\pi_0(\theta)$	Prior distribution
$f(y \mid \theta)$	Likelihood (intractable)
$\pi(\theta \mid y_{obs})$	Posterior approximation
$S(\cdot)$	Summary statistic function
s_{obs}	Observed summary
s_{sim}	Simulated summaries
$d(\cdot, \cdot)$	Distance function
ε	Tolerance threshold

Table 1: Core notation used throughout this report.

3 Choosing Summary Statistics

The ABC posterior defined in Section 2 depends critically on the choice of summary statistic $S(\cdot)$. If $S(\cdot)$ is sufficient for θ, replacing the full data with summaries introduces no additional approximation error beyond the tolerance ε. In practice, exact sufficiency is rarely achievable for complex models, and summary statistics must be designed such that they are *approximately* sufficient, capturing the information in the data most relevant to inferring θ while remaining low-dimensional enough for reliable distance comparison.

Two broad strategies exist. The first is to design statistics by hand, drawing on domain knowledge and mechanistic understanding of how each parameter affects the observable data. The second is to learn statistics systematically from pilot simulations, reducing the dependence on domain expertise. Both approaches are covered in this section. Hand-crafted statistics are discussed in

Section 3.1, and Semi-Automatic ABC, which automates statistic selection via a pilot regression, is discussed in Section 3.2.

A practical consideration common to both approaches is *dimension*. Adding summary statistics increases the dimensionality of the distance computation, which degrades the ABC approximation as accepted draws must be close to s_{obs} simultaneously in all dimensions, making acceptance increasingly rare. This is the curse of dimensionality in the ABC context. Statistics should therefore be included only when they carry information about at least one parameter that is not already captured by existing statistics.

3.1 Hand-Crafted Statistics

Hand-crafted summary statistics are functions of the data designed using knowledge of the model's structure. The central design challenge is *parameter separability*. Each parameter should ideally be the primary driver of at least one statistic, so that the distance function can distinguish between parameter values that produce similar aggregate behaviour through different mechanisms.

Design Principles

A useful starting point is to ask, for each parameter θ_k, if θ_k increases while all other parameters are held fixed, what changes in the observable data? Statistics that are sensitive to this change, and insensitive to changes in other parameters, are informative for θ_k specifically. When two parameters produce similar effects on a given data source, a phenomenon known as *parameter degeneracy*, statistics from an additional data source are required to break the degeneracy.

Redundancy and Dimensionality

Including redundant statistics, those that carry no information beyond what is already captured by existing statistics, increases dimensionality without improving the posterior concentration. In practice, redundancy can be assessed by comparing the ABC posterior with and without the candidate statistic. If the posterior is unchanged, the statistic is redundant and should be excluded.

3.2 Semi-Automatic ABC

Hand-crafted summary statistics rely entirely on domain knowledge and carry no optimality guarantee. Semi-Automatic ABC addresses this by deriving summary statistics from a pilot regression, reducing dimensionality to match the number of parameters while providing a principled optimality justification.

Theoretical Motivation

Let θ_0 denote the true parameter vector and $\hat{\theta}$ an estimate. The quality of $\hat{\theta}$ is measured by the weighted quadratic loss

$$L(\theta_0, \hat{\theta}) = (\theta_0 - \hat{\theta})^\top A\,(\theta_0 - \hat{\theta}),$$

where A is a positive definite matrix. Fearnhead and Prangle [2011] show that the minimal possible expected loss is achieved when $\hat{\theta} = E(\theta \mid \mathbf{y}_{obs})$, the posterior mean. Intuitively, the posterior mean retains exactly the information about θ contained in $\mathbf{y}$ and discards the rest, making it the ideal summary statistic. Since the posterior is intractable, $E[\theta \mid \mathbf{y}]$ cannot be computed directly. The pilot regression provides a tractable linear approximation.

Algorithm

Given parameters $\theta = (\theta_1, \ldots, \theta_p)$, Semi-Automatic ABC proceeds as follows:

1. **Pilot run.** For $i = 1, \ldots, N$: draw $\theta^{(i)} \sim \pi_0(\theta)$ and simulate $y^{(i)} \sim f(\cdot \mid \theta^{(i)})$.
2. **Candidate statistics.** Choose a transformation $h(\cdot)$ such that $h(\mathbf{y})$ is a vector of candidate summary statistics (hand-crafted features, moments, autocorrelations, or any other quantities computable from the raw data). The regression in the next step can only extract information already present in $h(\mathbf{y})$.
3. **Pilot regression.** For each parameter θ_j, fit the ordinary least squares model

$$\theta_j = \beta_0^{(j)} + \boldsymbol{\beta}^{(j)\top} h(\mathbf{y}) + \xi,$$

   using $\theta_j^{(1)}, \ldots, \theta_j^{(N)}$ as responses and $h(\mathbf{y}^{(1)}), \ldots, h(\mathbf{y}^{(N)})$ as covariates.
4. **Inference.** Run ABC using the p-dimensional summary vector whose jth component is

$$\hat{\beta}_0^{(j)} + \hat{\boldsymbol{\beta}}^{(j)\top} h(\mathbf{y}),$$

   the fitted value from the jth regression. This is the best linear approximation to $E[\theta_j \mid \mathbf{y}]$ and serves as a tractable surrogate for the theoretically optimal statistic. The resulting summary vector has dimension p, one component per parameter.

Limitations

The pilot regression assumes a linear relationship between $h(\mathbf{y})$ and each θ_j. When this relationship is strongly non-linear, the approximation to $E[\theta_j \mid \mathbf{y}]$ will be poor and the resulting statistics may perform no better than hand-crafted alternatives [Blum and François, 2009]. Furthermore, even under linearity, the quality of the derived statistics depends entirely on the informativeness of $h(\mathbf{y})$. If the candidate statistics carry little signal about θ, the regression learns nothing useful. Semi-automatic ABC therefore reduces but does not eliminate the need for domain knowledge in summary statistic design.

4 Sampling Algorithms

Given a summary statistic $S(\cdot)$, a distance function $d(\cdot, \cdot)$, and a tolerance ε, the task of posterior inference reduces to drawing samples from the ABC posterior

$$\pi(\theta \mid y_{obs}) \propto \pi_0(\theta) \cdot \mathbf{1}[d(S(y_{sim}), S(y_{obs})) \leq \varepsilon] \,.$$

The three sampling algorithms covered in this report differ in how they generate proposals and allocate the simulation budget. Rejection ABC draws proposals independently from the prior at a fixed tolerance. SMC-ABC evolves a population of particles through a sequence of decreasing tolerances, progressively refining the approximation. MCMC-ABC constructs a Markov chain that targets the ABC posterior directly, concentrating proposals in high-density regions.

Each algorithm involves a trade-off between simplicity, computational efficiency, and the quality of the posterior approximation. The decision criteria for choosing between them are discussed in Section 10.

4.1 Rejection ABC

Rejection ABC is the foundational sampling algorithm from which all other ABC methods derive. It is computationally straightforward and requires no tuning beyond the choice of tolerance, making it a natural starting point and a useful baseline for comparing more advanced methods.

Algorithm

Let $\pi_0(\theta)$ denote the prior, $f(\cdot \mid \theta)$ the simulator, and $S(\cdot)$ a fitted summary statistic. Given a tolerance $\varepsilon > 0$ and a simulation budget N, rejection ABC proceeds as follows:

1. For $i = 1, \ldots, N$: draw $\theta^{(i)} \sim \pi_0(\theta)$, simulate $y_{sim}^{(i)} \sim f(\cdot \mid \theta^{(i)})$, and compute $d^{(i)} = d(S(y_{sim}^{(i)}), S(y_{obs}))$.
2. Set $\varepsilon = Q_q\big(\{d^{(i)}\}_{i=1}^{N}\big)$, the q-quantile of the empirical distance distribution.
3. Return all $\theta^{(i)}$ for which $d^{(i)} \leq \varepsilon$.

The accepted draws form a sample from $\pi(\theta \mid y_{obs})$. Setting ε as a quantile of the empirical distances decouples the threshold from the absolute scale of the distance function and adapts automatically to the problem at hand.

Tolerance and the Bias-Efficiency Trade-off

The tolerance ε governs a fundamental trade-off. As $\varepsilon \to 0$, the ABC posterior converges to the true posterior, but the acceptance rate falls and a larger simulation budget N is required to obtain a fixed number of posterior samples. In the quantile formulation, the acceptance rate is fixed at q by construction, so tightening the posterior requires increasing N rather than decreasing q directly.

In practice, sensitivity to the choice of q should always be assessed by comparing posteriors at two or more values. A posterior that is stable across $q = 0.01$ and $q = 0.05$ gives greater confidence than one that shifts substantially.

Limitations

Rejection ABC has two well-known limitations that motivate the methods in subsequent sections.

1. *Tolerance-induced bias*: Simulated summaries are close to but not identical to s_{obs}, leaving residual spread in the posterior that does not reflect genuine parameter uncertainty.
2. *Computation wastage*: Proposals are drawn uniformly from the prior regardless of where the posterior mass lies. At small tolerances, the vast majority of draws are rejected, and efficiency degrades as ε tightens.

Regression adjustment (Section 5.1) addresses the first limitation as a post-processing correction. SMC-ABC (Section 4.2) and MCMC-ABC (Section 4.3) address the second by concentrating proposals near the posterior.

Implementation

In `abclib`, rejection ABC is implemented in `Rejection`ABC (inheriting from `BaseSampler`). The `sample` method accepts `s_obs`, `n_simulations`, and a quantile `q`. It builds a reference table of all N distances, sets ε at the q-quantile via `select_epsilon`, and returns all accepted draws as an ABC`Result`. The summary statistic must be fitted on pilot simulations before being passed to the sampler.

4.2 SMC-ABC

Rejection ABC draws proposals uniformly from the prior regardless of where the posterior mass lies, wasting the majority of its simulation budget on regions of low posterior density. SMC-ABC improves upon this by concentrating proposals near the posterior, refining the sampling region through a sequence of successively tighter tolerances.

Algorithm

SMC-ABC evolves a population of M particles through tolerances $\varepsilon_1 > \varepsilon_2 > \cdots > \varepsilon_T$, concentrating simulation effort progressively closer to the posterior.

1. **Stage 1.** For $i = 1, \ldots, M$: draw $\theta^{(i)} \sim \pi_0(\theta)$, simulate $y_{\text{sim}}^{(i)} \sim f(\cdot \mid \theta^{(i)})$, compute $d^{(i)} = d\Big(S(y_{\text{sim}}^{(i)}),\ S(y_{\text{obs}})\Big)$, and retain if $d^{(i)} \le \varepsilon_1$, else repeat. Assign uniform weights $w^{(i)} = 1/M$.
2. **Stages** $2, \ldots, T$**.** At each stage t:
   (a) **Perturb.** Sample θ^* from the previous population with probability proportional to $w^{(j)}$, then apply a Gaussian perturbation kernel $\theta^* \sim \mathcal{N}(\theta^{(j)}, H)$, where H is a diagonal matrix with entries

   $$h_k = 2\left(\sum_j w^{(j)}\left(\theta_k^{(j)} - \bar{\theta}_k\right)^2\right)^{\frac{1}{2}},$$

   and $\bar{\theta}_k$ is the weighted mean of the kth parameter. The adaptive bandwidth ensures the kernel naturally contracts as the posterior concentrates, preventing over-dispersed proposals in later stages [Beaumont et al., 2009].

   (b) **Retain.** Simulate $y_{\text{sim}}^* \sim f(\cdot \mid \theta^*)$ and retain θ^* if $d(S(y_{\text{sim}}^*),\ S(y_{\text{obs}})) \le \varepsilon_t$, else repeat from (a).

(c) **Weight.** Assign importance weight

$$w^* \propto \frac{\pi(\theta^*)}{\sum_j w^{(j)} K(\theta^* \mid \theta^{(j)})},$$

where the denominator is the mixture density from which θ^* was effectively drawn. The ratio corrects for the fact that proposals no longer come from the prior.

(d) **Resample.** If the effective sample size

$$\mathrm{ESS} = \left(\sum_j \left(w^{(j)} \right)^2 \right)^{-1} < \frac{M}{2},$$

resample M particles with replacement proportional to their weights and reset weights to $1/M$.

(e) **Update tolerance.** Set ε_{t+1} to the median of the accepted distances from stage t. This ensures the tolerance decreases at a rate the particle population can sustain.

Limitations

SMC-ABC has two main limitations:

1. *Particle degeneracy*: If the tolerance decreases too quickly, most particles are rejected and the ESS collapses, with a small number of particles carrying almost all the weight. This causes the posterior approximation to degrade.
2. *Tuning complexity*: SMC-ABC requires additional design choices that rejection ABC does not, namely the number of particles M, the initial tolerance ε_1, and the kernel type.

Implementation

SMCABC extends `BaseSampler` with an additional `prior_density` argument – a vectorised callable returning prior densities for a batch of particles, required for importance weight computation. The `sample` method accepts `M` (particles), `T` (stages), and `q` (quantile for adaptive tolerance). The tolerance sequence is stored in `ABCResult.epsilons`.

4.3 MCMC-ABC

Rejection ABC draws proposals independently from the prior at every step, wasting most of its simulation budget on low-density regions. MCMC-ABC addresses this by constructing a Markov chain whose stationary distribution is the ABC posterior $\pi_{ABC}(\theta \mid y_{obs})$, concentrating proposals in high-posterior-density regions by conditioning each move on the previously accepted parameter value.

Theoretical Justification

MCMC-ABC is an application of the Metropolis-Hastings algorithm to the ABC posterior. In standard Metropolis-Hastings the acceptance ratio involves the likelihood $f(y_{obs} \mid \theta)$, which is intractable. MCMC-ABC replaces the likelihood with the ABC indicator: a draw θ is treated as consistent with the data if and only if $d(S(y_{sim}), S(y_{obs})) \leq \varepsilon$.

The key observation is that if the current state θ has already passed the ε-gate, and the proposed state θ' also passes it, then both are assigned equal weight by the ABC likelihood approximation. The ratio of ABC likelihoods is therefore $\mathbf{1}/\mathbf{1} = 1$ and cancels from the acceptance ratio, leaving only the prior ratio and the proposal ratio:

$$\alpha(\theta, \theta') = \min\left\{1, \frac{\pi_0(\theta')\, q(\theta \mid \theta')}{\pi_0(\theta)\, q(\theta' \mid \theta)}\right\}.$$

This construction satisfies detailed balance with respect to $\pi_{ABC}(\theta \mid y_{obs})$, guaranteeing it as the unique stationary distribution of the chain [Marjoram et al., 2003].

Algorithm

The algorithm initialises from an accepted draw and then runs a Metropolis-Hastings loop with an ε-gate as a pre-filter.

1. **Initialisation.** Draw $\theta^{(1)} \sim \pi_0(\theta)$, simulate $y_{sim}^{(1)} \sim f(\cdot \mid \theta^{(1)})$, and compute $d^{(1)} = d(S(y_{sim}^{(1)}), S(y_{obs}))$. If $d^{(1)} \leq \varepsilon$ proceed, otherwise repeat until an accepted draw is found.
2. **Iteration $i = 2, \ldots, N$.** Given the current state $\theta^{(i-1)}$:
   (a) **Propose.** Draw $\theta' \sim q(\cdot \mid \theta^{(i-1)})$ from a local proposal kernel.
   (b) **Simulate and gate.** Simulate $y'_{sim} \sim f(\cdot \mid \theta')$ and compute $d' = d(S(y'_{sim}), S(y_{obs}))$. If $d' > \varepsilon$, reject immediately and set $\theta^{(i)} = \theta^{(i-1)}$; return to (a).
   (c) **MH accept/reject.** If $d' \leq \varepsilon$, accept θ' with probability
   $$\alpha(\theta^{(i-1)}, \theta') = \min\left\{1, \frac{\pi_0(\theta')\, q(\theta^{(i-1)} \mid \theta')}{\pi_0(\theta^{(i-1)})\, q(\theta' \mid \theta^{(i-1)})}\right\}.$$
   Set $\theta^{(i)} = \theta'$ if accepted, else $\theta^{(i)} = \theta^{(i-1)}$.

A common choice for q is an isotropic Gaussian centred at the current state, $q(\theta' \mid \theta) = \mathcal{N}(\theta, \sigma^2 I)$, for which the proposal ratio cancels and the acceptance probability reduces to the prior ratio alone.

Limitations

MCMC-ABC has three practical limitations.

1. *Poor mixing.* If the proposal kernel is poorly tuned, the chain moves slowly through the parameter space and produces highly correlated samples. The chain's stationary distribution remains π_{ABC}, but the effective sample size (ESS) can be far smaller than the chain length N, meaning the information content of the output is much less than the simulation budget suggests. Unlike rejection ABC, where samples are independent by construction, correlation is an unavoidable baseline cost in MCMC-ABC that must always be diagnosed.
2. *Local trapping.* When the ABC posterior is multimodal, a local proposal kernel can become trapped in a single mode, producing a sample that misrepresents the full posterior. SMC-ABC is better suited to multimodal targets because its particle population can simultaneously cover multiple modes.
3. *Proposal tuning.* The step size σ of the proposal kernel must be tuned manually. A step size that is too large causes most proposals to fail the ε-gate, while one that is too small leads to poor mixing. Neither problem has a built-in remedy, in contrast to SMC-ABC where the bandwidth adapts to the particle population.

Implementation

MCMCABC extends `BaseSampler` with `prior_pdf` (a scalar density callable) and `proposal_std`. The `sample` method accepts `n_samples` and a fixed `epsilon`. Each chain step first applies the ε-gate; only proposals that pass proceed to the MH ratio. For a symmetric Gaussian proposal the ratio reduces to the prior ratio alone. Every chain state is recorded, including repeated states from rejected proposals, preserving the true correlation structure of the chain.

5 Post-Processing

5.1 Regression Adjustment

As mentioned in Section 4.1, accepted draws have $s_{sim} \neq s_{obs}$ because of the tolerance $\varepsilon > 0$. This mismatch between observed and simulated summaries results in a tolerance-induced bias for the accepted draws. Regression adjustment addresses this limitation by adjusting the parameters, thereby reducing the discrepancy between simulated and observed statistics.

Algorithm

We fit an OLS linear model following:

$$\theta^{(i)} = a + B(s^{(i)} - s_{obs}) + \xi$$

where $\theta^{(i)}$ and $s^{(i)}$ are the ith accepted draw and its associated summary vector. We fit the model on the standardised summaries so that statistics on larger scales do not dominate the coefficient estimates. Then a is an intercept vector and B is a matrix of regression coefficients. The model is expressed as a deviation from s_{obs}, such that the predicted value at $s^{(i)} = s_{obs}$ is exactly a. The adjusted sample is then defined by:

$$\theta^{(i)}_{adj} = \theta^{(i)} - \hat{B}(s^{(i)} - s_{obs})$$

This method has the advantage of improving the quality of the posterior whilst being computationally cheap, as no additional simulations are required beyond those already generated for the rejection ABC. The adjustment shifts each accepted $\theta^{(i)}$ by the component of the deviation $s^{(i)} - s_{obs}$ explained by the linear model.

Limitations

The validity of the regression is dependent on two assumptions:

1. *Homoscedasticity*: If the variance of ξ is dependent on $s^{(i)}$, the correction will be over- or under- applied in different regions.
2. *Linearity*: If the true relationship between θ and s is non-linear, the linear correction will introduce its own bias. Blum and François [2009] proposes non-linear extensions.

Additionally, adjusted samples can fall outside the prior support, and can be handled either through truncation or reflection. Truncation has the advantage of being simple and quick, but can introduce bias near boundaries by discarding or cutting off adjusted values that fall outside the permitted range. Reflection bounces adjusted values back into the valid region, preserving the density shape better than truncation.

Implementation

`RegressionAdjustment` takes `prior_bounds` as a list of `(lower, upper)` tuples. `fit` standardises the summary deviations $(s^{(i)} - s_{obs})$ and fits one OLS model per parameter. `adjust` applies the correction and reflects out-of-bounds samples iteratively back into the prior support, with a maximum of ten bounces before clipping.

6 An Alternative Fork: Synthetic Likelihood

While the ABC methods previously discussed are non-parametric, making no assumption about the distribution of summaries and using a distance threshold instead, Synthetic Likelihood is parametric and replaces the intractable likelihood with a Gaussian approximation and then embeds this within a Metropolis-Hastings sampler [Price et al., 2018, Wood, 2010].

Theoretical Justification

For a proposed parameter θ, the summary statistics of repeated simulations $s_{sim}^{(1)}, \ldots, s_{sim}^{(M)}$ form a distribution whose shape depends on θ. The Central Limit Theorem provides justification for approximating this distribution as Gaussian. As the length n of each simulated dataset grows, sample summary statistics (means, autocorrelations, and similar quantities) converge in distribution to a Gaussian by CLT. Under this approximation, the summary statistics follow

$$s_{sim} \mid \theta \;\sim\; \mathcal{N}(\mu(\theta),\, \Sigma(\theta))\,,$$

where $\mu(\theta) = E[s_{sim} \mid \theta]$ and $\Sigma(\theta) = \mathrm{Var}[s_{sim} \mid \theta]$. Since $\mu(\theta)$ and $\Sigma(\theta)$ are unknown, they are estimated from M simulations at each proposed θ:

$$\mu_M(\theta) = \frac{1}{M}\sum_{j=1}^{M} s_{sim}^{(j)}, \qquad \Sigma_M(\theta) = \frac{1}{M-1}\sum_{j=1}^{M}\big(s_{sim}^{(j)} - \mu_M(\theta)\big)\big(s_{sim}^{(j)} - \mu_M(\theta)\big)^{\top}.$$

The synthetic likelihood is then defined as the Gaussian density evaluated at the observed summaries s_{obs}:

$$\hat{L}_{SL}(\theta) = \mathcal{N}(s_{obs};\; \mu_M(\theta),\, \Sigma_M(\theta))\,.$$

This replaces the intractable $f(s_{obs} \mid \theta)$ in the posterior:

$$\pi_{SL}(\theta \mid s_{obs}) \;\propto\; \pi_0(\theta)\; \hat{L}_{SL}(\theta).$$

Algorithm

Synthetic Likelihood is embedded within a Metropolis-Hastings loop. At each step, M simulations are run at the proposed θ' to evaluate $\hat{L}_{SL}(\theta')$, which enters the acceptance ratio alongside the stored value from the current state.

1. **Initialisation.** Draw $\theta^{(1)} \sim \pi_0(\theta)$. Run M simulations $y_{sim}^{(j)} \sim f(\cdot \mid \theta^{(1)})$, compute $s_{sim}^{(j)} = S(y_{sim}^{(j)})$ for $j = 1, \dots, M$, and evaluate $\hat{L}_{SL}(\theta^{(1)})$.
2. **Iteration** $i = 2, \dots, N$**.** Given current state $\theta^{(i-1)}$ and its synthetic likelihood $\hat{L}_{SL}(\theta^{(i-1)})$:
   (a) **Propose.** Draw $\theta' \sim q(\cdot \mid \theta^{(i-1)})$ from a local proposal kernel.
   (b) **Simulate.** For $j = 1, \dots, M$, simulate $y_{sim}^{(j)} \sim f(\cdot \mid \theta')$ and compute $s_{sim}^{(j)} = S(y_{sim}^{(j)})$.
   (c) **Estimate.** Compute $\mu_M(\theta')$ and $\Sigma_M(\theta')$ from the M simulated summaries.
   (d) **Evaluate.** Compute the synthetic likelihood

   $$\hat{L}_{SL}(\theta') = \mathcal{N}(s_{obs};\ \mu_M(\theta'),\ \Sigma_M(\theta'))\,.$$

   (e) **Accept/reject.** Compute the Metropolis-Hastings acceptance probability

   $$\alpha(\theta^{(i-1)}, \theta') = \min\left\{1,\ \frac{\pi_0(\theta')\,\hat{L}_{SL}(\theta')\,q(\theta^{(i-1)} \mid \theta')}{\pi_0(\theta^{(i-1)})\,\hat{L}_{SL}(\theta^{(i-1)})\,q(\theta' \mid \theta^{(i-1)})}\right\}.$$

   Set $\theta^{(i)} = \theta'$ with probability α, otherwise set $\theta^{(i)} = \theta^{(i-1)}$.

Unlike MCMC-ABC where the ABC likelihood terms cancelled from the acceptance ratio because both states passed the ε-gate equally, the synthetic likelihoods $\hat{L}_{SL}(\theta')$ and $\hat{L}_{SL}(\theta^{(i-1)})$ are continuous values and do not cancel. Their ratio is what drives the chain toward high-likelihood regions.

Limitations

Synthetic Likelihood has three practical limitations.

1. *Gaussian assumption failure.* If the true distribution of summary statistics is non-Gaussian, the synthetic likelihood is a poor approximation and the resulting posterior will be misleading. Unlike ABC, where a poor summary statistic merely widens the posterior, Synthetic Likelihood has no tolerance parameter to absorb this error: a misspecified Gaussian fit produces a systematically biased posterior with no internal signal that anything is wrong.

2. *Computational cost.* Each MCMC step requires M simulator calls to estimate $\mu_M(\theta')$ and $\Sigma_M(\theta')$. Larger M produces more stable estimates but multiplies the total simulation budget by M relative to MCMC-ABC, which requires only one simulation per step.

3. *Sensitivity to summary statistic choice.* The quality of the Gaussian approximation depends on the chosen summaries. Summaries that are themselves highly non-Gaussian, or that are near-redundant, will produce an unstable or poorly conditioned $\Sigma_M(\theta)$, degrading the approximation regardless of M.

Implementation

`SyntheticLikelihood` is implemented as a standalone class (not inheriting from `BaseSampler`, since it does not use a distance function or return an `ABCResult`). It accepts `prior_pdf` and `proposal_std` alongside the standard prior, simulator, and summary statistic arguments. The `_log_likelihood` method draws M replicates, estimates μ_M and Σ_M, and evaluates the Gaussian log-density at s_{obs} using `scipy.stats.multivariate_normal.logpdf`. Working in log space is essential for numerical stability: the synthetic likelihood can span many orders of magnitude across the parameter space, causing floating-point underflow to zero if evaluated in probability space. The MH acceptance ratio is computed entirely in log space, and the stored synthetic log likelihood from the current state is reused at each step without additional simulations. Results are returned as `SLResult`, which stores log likelihoods and exposes `posterior_mean`, `credible_interval`, and `acceptance_rate`.

7 Validation

ABC methods use simulations to approximate an unknown likelihood function. Without validation, it is difficult to ascertain whether the approximate posterior reflects genuine parameter uncertainty or merely the deficiencies of the model, the chosen summary statistics, or the tolerance threshold. The three methods below are complementary: SBC checks global calibration of the posterior, PPC checks whether the model can reproduce observed data features, and STR checks instance-level recovery at specific parameter values. A method that passes all three gives substantially stronger guarantees than one that passes any single diagnostic. These diagnostics are applied to the MA(2) model in Section 8 and the Lotka-Volterra model in Section 9.

7.1 Simulation-Based Calibration (SBC)

SBC tests whether the posterior is correctly calibrated by exploiting a theoretical property of well-specified Bayesian inference. If $\theta^* \sim \pi_0(\theta)$ and $\theta^{(1)}, \ldots, \theta^{(L)} \sim \pi(\theta \mid y^*)$ are posterior draws obtained by running inference on data simulated from θ^*, then the rank

$$r = \sum_{l=1}^{L} \mathbf{1}\left[\theta^{(l)} < \theta^*\right]$$

is uniform on $\{0, \ldots, L\}$ if and only if the posterior is correctly calibrated [Talts et al., 2020]. Repeating this procedure over many draws $\theta^{(i)} \sim \pi_0(\theta)$ and plotting the resulting rank histogram provides a diagnostic of the full inference pipeline. Departures from uniformity carry specific interpretations: a U-shaped histogram indicates an over-dispersed posterior; a hill-shaped histogram indicates an under-dispersed posterior; a skewed histogram indicates a systematic bias in one direction.

Algorithm

1. **Sample.** For $i = 1, \ldots, N$, draw $\theta^{(i)} \sim \pi_0(\theta)$ and simulate $y^{(i)} \sim f(\cdot \mid \theta^{(i)})$.
2. **Infer.** Run the ABC algorithm on each $y^{(i)}$ to obtain L posterior draws from $\pi(\theta \mid y^{(i)})$.
3. **Rank.** For each i, compute the rank $r^{(i)}$ of $\theta^{(i)}$ within its posterior sample. Under a correctly calibrated posterior, the ranks $\{r^{(i)}\}$ are uniformly distributed on $\{0, \ldots, L\}$.
4. **Assess.** Plot the rank histogram and conduct a Kolmogorov-Smirnov test against the uniform distribution. The KS statistic is the maximum absolute difference between the empirical CDF of the ranks and the uniform CDF. A p-value below 0.05 provides evidence against calibration; a p-value above 0.05 fails to reject calibration but does not confirm it.

Limitations

SBC tests calibration of the full pipeline jointly but cannot identify which component is responsible for a failure. A poor summary statistic, an inappropriate tolerance, or a sampler bug will all produce non-uniform ranks without distinguishing themselves. SBC is also computationally expensive, requiring N full inference runs.

7.2 Posterior Predictive Checks (PPC)

PPC asks whether data simulated under the approximate posterior resembles the observed data. Unlike SBC, which checks calibration globally, PPC checks whether the model is adequate for the specific observed dataset. For each posterior draw $\theta^{(i)} \sim \pi_{ABC}(\theta \mid y_{obs})$, new replicate data $y_{rep}^{(i)} \sim f(\cdot \mid \theta^{(i)})$ is simulated and compared to y_{obs} via a test statistic $T(\cdot)$.

Algorithm

1. **Simulate.** For $i = 1, \ldots, N$: draw $\theta^{(i)} \sim \pi_{ABC}(\theta \mid y_{obs})$ and simulate $y_{rep}^{(i)} \sim f(\cdot \mid \theta^{(i)})$.

2. **Test statistic.** Choose $T(\cdot)$ to capture a feature of the data the model is supposed to explain. Compute $T(y_{rep}^{(i)})$ for each replicate and $T(y_{obs})$ for the observed data.

3. **Posterior predictive p-value.** Compute

$$p = P(T(y_{rep}) \geq T(y_{obs})) \approx \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}\left[T(y_{rep}^{(i)}) \geq T(y_{obs})\right].$$

   Values of p near 0 or 1 indicate that the observed data is extreme relative to the posterior predictive distribution, pointing to model misfit. Values near 0.5 indicate consistency with the observed data under T, but are not proof of correctness.

Limitations

The diagnostic is only as informative as the choice of $T(\cdot)$: a statistic insensitive to model misfit will produce uninformative p-values even when the model is wrong. Additionally, PPC uses the data twice, once to fit the posterior and once to evaluate it, making it conservative. A model can pass PPC while being wrong in ways that $T(\cdot)$ does not capture.

7.3 Synthetic Truth Recovery (STR)

STR checks whether the approximate posterior recovers a known ground truth parameter θ^* used to generate synthetic data. Unlike SBC, which checks calibration averaged over the prior, STR is an instance-level check that can expose region-specific failures. A sampler may be well-calibrated on average but fail systematically near the boundaries of the prior support, which SBC would average over but a grid of STR runs would detect.

Algorithm

1. **Ground truth.** Choose θ^* and simulate $y_{obs}^* \sim f(\cdot \mid \theta^*)$.

2. **Infer.** Run the ABC algorithm with y_{obs}^* as observed data to obtain the approximate posterior $\pi_{ABC}(\theta \mid y_{obs}^*)$.

3. **Check coverage.** Compute the 90% credible interval of the approximate posterior and check whether θ^* falls within it.

4. **Repeat.** Repeat steps 1–3 across a grid of θ^* values spanning the prior support and report the empirical coverage. Coverage substantially below 90% indicates systematic recovery failure.

Limitations

STR is only informative at the tested θ^* values. A single run proves very little, meaningful conclusions require a grid spanning the prior support. Even then, STR checks recovery but not calibration. A posterior can contain θ^* within its credible interval while still being poorly shaped, which SBC would detect but STR would not.

8 Validation Model: MA(2)

To ensure that each implementation is correct, we make use of a tractable MA(2) model whose exact posterior is available in closed form, allowing direct comparison with the ABC posterior. Every method implemented in `abclib` is validated on this model before application to the Lotka-Volterra case study. The triangular posterior shape tests whether each sampler correctly handles constrained parameter spaces, and convergence to the exact posterior as $\varepsilon \to 0$ confirms both correct implementation and that the chosen summary statistics are sufficiently informative.

8.1 Model Definition

Motivated by financial returns, each observation is modelled as a weighted combination of two past random shocks:

$$y_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2}, \quad \varepsilon_t \overset{iid}{\sim} \mathcal{N}(0, 1).$$

To ensure invertibility, the parameters must satisfy:

$$\theta_1 + \theta_2 < 1, \quad \theta_2 - \theta_1 < 1, \quad -1 < \theta_2 < 1.$$

Without these constraints, different parameter values produce identical auto-covariance structures and the parameters cannot be identified. The prior is taken to be uniform over the invertibility region $\mathcal{R}$.

8.2 Exact Posterior

The likelihood for an observed series $y_1, \ldots, y_T$ given θ is Gaussian. The exact posterior is proportional to the likelihood times the uniform prior, restricted to $\mathcal{R}$:

$$\pi(\theta \mid y) \propto \mathcal{N}(y;\, 0,\, \Sigma(\theta)) \cdot \mathbf{1}[\theta \in \mathcal{R}],$$

where $\Sigma(\theta)$ is the auto-covariance matrix of the MA(2) process. This matrix has a known closed form, where the (i, j)th entry depends only on $|i - j|$ and the MA coefficients (θ_1, θ_2). This exact posterior serves as ground truth against which all ABC posteriors are compared.

8.3 Summary Statistics

The natural summary statistics are the lag-1 and lag-2 sample autocorrelations:

$$\hat{\rho}_k = \frac{\displaystyle\sum_{t=k+1}^{T} (y_t - \bar{y})(y_{t-k} - \bar{y})}{\displaystyle\sum_{t=1}^{T} (y_t - \bar{y})^2}, \quad k = 1, 2.$$

The theoretical autocorrelations of an MA(2) process are direct functions of θ_1 and θ_2, making $(\hat{\rho}_1, \hat{\rho}_2)$ approximately sufficient statistics for this model. As $\varepsilon \to 0$, the ABC posterior using these statistics should converge to the exact posterior, providing a direct test of correctness for each implemented method.

8.4 Validation Results

All methods implemented in `abclib` were validated on the MA(2) model with true parameters $\theta^* = (0.6, 0.2)$ and a time series of length $T = 100$. Summary statistics are the lag-1 and lag-2 sample autocorrelations, fitted on 2000 pilot simulations. Figure 1 shows the ABC posterior samples from each method overlaid on the exact posterior contours.

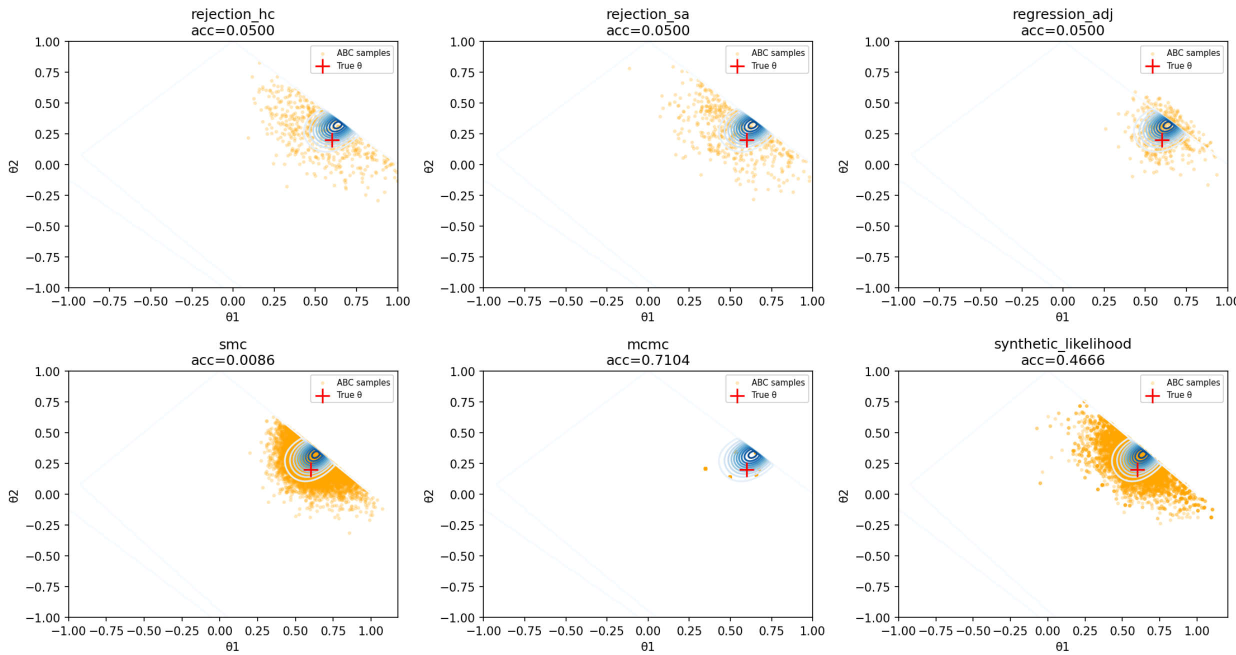

Figure 1: ABC posterior samples (orange) from each method overlaid on exact posterior contours (blue). True parameter $\theta^* = (0.6, 0.2)$ marked with a red cross. Rejection ABC (HC and SA) use $N = 10{,}000$ simulations at $q = 0.05$. SMC-ABC uses $M = 10{,}000$ particles over $T = 5$ stages. MCMC-ABC runs a chain of 10,000 steps at $\varepsilon = 0.05$. Regression adjustment is applied to the rejection ABC (HC) result. Synthetic likelihood uses $N = 10{,}000$ steps with $M = 100$ replicates per step.

Table 2 summarises posterior means and 90% credible intervals for each method.

Method	$\hat{\theta}_1$	$\hat{\theta}_2$	**90% CI** (θ_1)	**90% CI** (θ_2)
Rejection ABC (HC)	0.527	0.250	[0.233, 0.865]	[-0.064, 0.582]
Rejection ABC (SA)	0.522	0.262	[0.214, 0.846]	[-0.058, 0.568]
Regression Adj.	0.618	0.257	[0.456, 0.795]	[0.045, 0.462]
SMC-ABC	0.604	0.262	[0.438, 0.787]	[0.060, 0.437]
MCMC-ABC	0.628	0.259	[0.497, 0.705]	[0.142, 0.383]
Synthetic Likelihood	0.589	0.270	[0.372, 0.793]	[0.037, 0.476]
True θ^*	0.600	0.200	—	—

Table 2: Posterior means and 90% credible intervals for each method on the MA(2) model with $\theta^* = (0.6, 0.2)$.

All methods recover θ_1 well. SMC-ABC and MCMC-ABC produce notably tighter credible intervals than rejection ABC, reflecting their more efficient concentration of proposals near the posterior. The credible intervals for θ_2 are wider across all methods, consistent with the weaker identifiability of θ_2 through the lag-2 autocorrelation given the non-linear relationship between (θ_1, θ_2) and the autocorrelation structure.

SBC Results

SBC was run over 500 trials with $L = 100$ posterior draws per trial, first on raw rejection ABC and then on the regression-adjusted result. Figure 2 shows the rank histograms for the rejection ABC posterior.

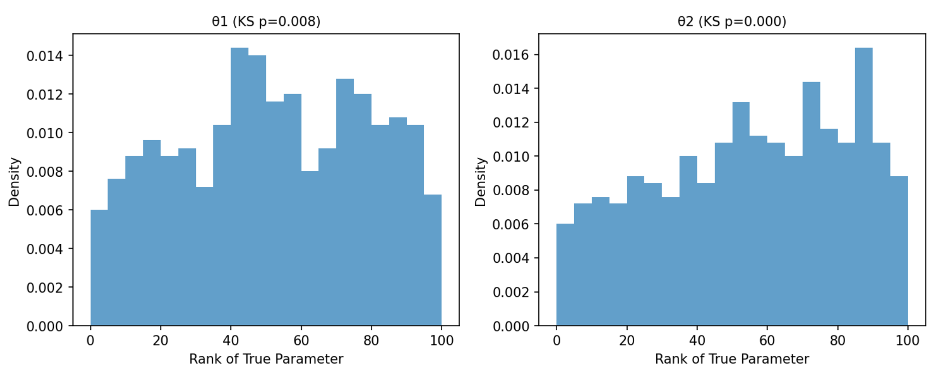

Figure 2: SBC rank histograms for θ_1 (top) and θ_2 (bottom) under rejection ABC at $q = 0.05$.

The rank histogram for θ_2 shows a clear right-skew (KS p ≈ 0), indicating that the rejection ABC posterior systematically underestimates θ_2. This is consistent with tolerance-induced bias: at finite ε, the non-linear relationship between the lag-2 autocorrelation and (θ_1, θ_2) causes residual summary spread to bias the marginal posterior for θ_2 downward. The mild left-skew in θ_1 (KS p=0.008) suggests a corresponding upward bias, consistent with the positive correlation between the two parameters in the posterior.

Figure 3 shows the rank histograms after regression adjustment.

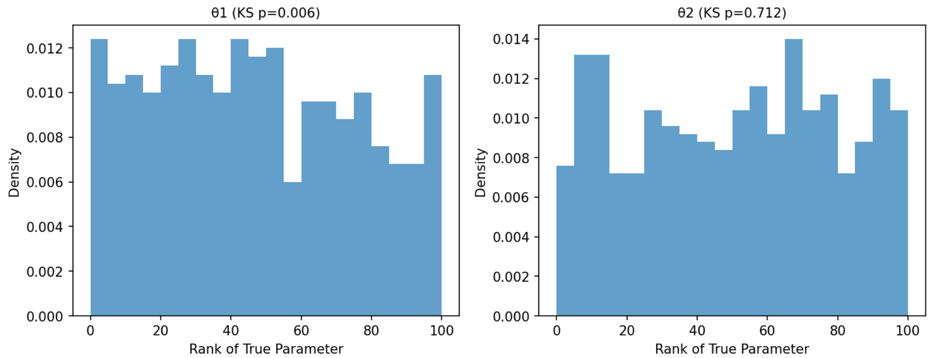

Figure 3: SBC rank histograms for θ_1 (top) and θ_2 (bottom) after regression adjustment.

Regression adjustment substantially corrects the θ_2 bias: the KS p-value increases from ≈ 0 to 0.712, and the rank histogram is visually uniform. This confirms the bias was tolerance-induced rather than a consequence of summary statistic insufficiency. The θ_1 miscalibration is largely unchanged (KS p=0.006), consistent with the linearity limitation of regression adjustment discussed in Section 5.1: the cross-parameter regression coefficients correct θ_2 but cannot simultaneously correct θ_1 when the true relationship is non-linear. This illustrates that regression adjustment should be treated as an approximation and its effect always validated with SBC.

STR Results

STR was run across a grid of 30 parameter values spanning the invertibility region. Figure 4 shows the credible intervals for each grid point.

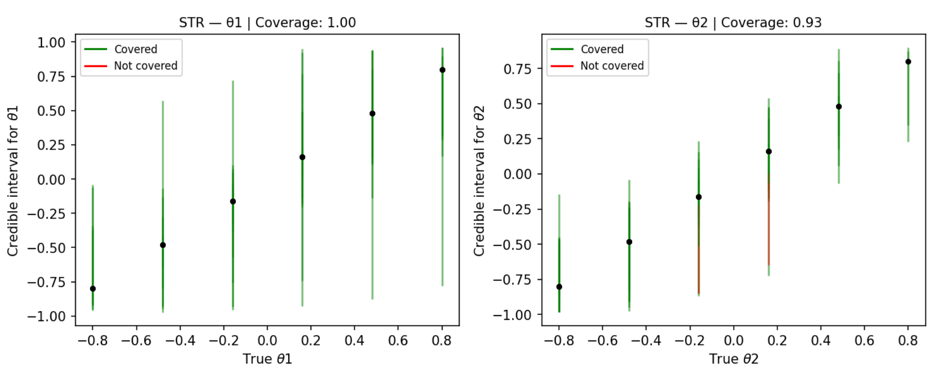

Figure 4: STR credible intervals across the parameter grid for θ_1 (top) and θ_2 (bottom). Green intervals contain the true parameter; red intervals do not.

Empirical coverage is 1.00 for θ_1 and 0.93 for θ_2, both at or above the nominal 90% credible mass. The elevated coverage for θ_1 reflects the mild overdispersion visible in the wide rejection ABC intervals in Figure 1. The instance-level recovery for θ_2 is acceptable despite the population-level calibration failure detected by SBC, illustrating the complementarity of the two diagnostics: SBC detects systematic bias averaged over the prior, while STR checks whether individual credible intervals are wide enough to contain the truth at specific parameter values. A biased but overdispersed posterior can pass STR while failing SBC.

PPC Results

PPC was run on the regression-adjusted posterior using sample mean and sample variance as test statistics. The posterior predictive p-values were 0.260 (mean) and 0.273 (variance), both well within the acceptable range. The model reproduces the mean and variance of the observed time series under the approximate posterior, providing no evidence of model misfit for these statistics.

9 Case Study: Lotka-Volterra

The Lotka-Volterra model describes the population dynamics of two interacting species: a prey population x and a predator population y. It is widely used in ecology to model systems such as lynx and hare, or parasites and hosts. The model is intractable for two reasons. First, the stochastic version is a continuous-time Markov chain whose transition density satisfies the Chapman–Kolmogorov equations over a state space of size $O(x_{max} \times y_{max})$, making exact likelihood evaluation computationally infeasible as population sizes grow. Second, log-normal observation noise is layered over the latent dynamics, adding a further integral over the latent trajectory that has no closed form. Both sources of intractability prevent direct likelihood evaluation, making Lotka-Volterra a natural benchmark for ABC methods.

9.1 Model Definition

The model ignores environmental limits such as carrying capacity and time delays. The prey and predator populations evolve according to:

$$\frac{dx}{dt} = \alpha x - \beta xy, \qquad \frac{dy}{dt} = \delta xy - \gamma y,$$

where α is the prey intrinsic growth rate, β is the predation rate, δ is the predator reproduction efficiency, and γ is the predator mortality rate. The two populations form a periodic oscillation around the non-zero stationary point:

$$x^* = \frac{\gamma}{\delta}, \qquad y^* = \frac{\alpha}{\beta}.$$

The effect of each parameter on the long-run population averages follows directly from these expressions:

- *Prey growth rate* α: Increasing α raises y^*, as prey recovers faster between predation events, sustaining a larger predator population on average.
- *Predation rate* β: Increasing β lowers y^*, as predators deplete their food source more rapidly, suppressing their own long-run average.
- *Predator reproduction efficiency* δ: Increasing δ lowers x^*, as predators convert prey into offspring more efficiently, keeping prey suppressed at a lower average level.
- *Predator mortality rate* γ: Increasing γ raises x^*, as predators die faster and exert less hunting pressure on prey.

9.2 Stochastic Simulator

The deterministic Lotka-Volterra ODEs do not constitute a valid simulator for ABC inference, because a deterministic model produces identical output for any fixed θ and initial condition, making the distance $d(S(y_{sim}), S(y_{obs}))$ a deterministic function of θ rather than a stochastic one. ABC requires a stochastic simulator so that the ABC posterior reflects genuine parameter uncertainty rather than a point mass at the parameter value that minimises the distance.

Observation Model

We introduce stochasticity through log-normal observation noise layered over the deterministic ODE trajectory. At each time step t, the observed prey and predator counts are:

$$\tilde{x}_t = x_t \exp(\epsilon_t^x), \qquad \tilde{y}_t = y_t \exp(\epsilon_t^y), \qquad \epsilon_t^x, \epsilon_t^y \overset{iid}{\sim} \mathcal{N}(0, \sigma^2),$$

where x_t and y_t are the latent ODE states at time t and σ controls the noise level. Log-normal noise is standard for population count data, as it preserves positivity and scales multiplicatively with population size [Wood, 2010]. This observation model is the primary source of intractability. The likelihood requires integrating over the latent trajectory $(x_1, y_1), \ldots, (x_T, y_T)$, which has no closed form.

Numerical Integration

The latent ODE trajectory is integrated using the Euler-Maruyama method with step size Δt:

$$x_{t+1} = x_t + (\alpha x_t - \beta x_t y_t)\,\Delta t, \qquad y_{t+1} = y_t + (\delta x_t y_t - \gamma y_t)\,\Delta t.$$

Populations are clipped at 10^{-6} after each step to prevent numerical extinction. The simulator is initialised at (x_0, y_0) and run for T steps, returning the noisy observations $\{(\tilde{x}_t, \tilde{y}_t)\}_{t=1}^T$.

Prior and Initial Conditions

The prior is taken to be independent uniform distributions over ecologically plausible parameter ranges:

$$\alpha \sim \mathcal{U}(0.1, 1.0), \quad \beta \sim \mathcal{U}(0.01, 0.5), \quad \delta \sim \mathcal{U}(0.01, 0.5), \quad \gamma \sim \mathcal{U}(0.1, 1.0).$$

The simulator is initialised at $x_0 = 50$ prey and $y_0 = 20$ predators, with step size $\Delta t = 0.1$ and observation noise $\sigma = 0.1$. These values produce stable oscillatory dynamics across the prior support while keeping population sizes in a numerically tractable range.

True Parameters

The case study uses true parameters $\theta^* = (\alpha, \beta, \delta, \gamma) = (0.5, 0.1, 0.1, 0.5)$, which produce a stable predator–prey cycle with period of approximately 10 time units. The observed dataset is a single trajectory of length $T = 100$ simulated under θ^* with log-normal observation noise.

9.3 Summary Statistics

Population counts in Lotka-Volterra span several orders of magnitude, so all summary statistics are computed on log-transformed populations following Wood [2010]. Table 3 lists the statistics used and their primary parameter sensitivity.

Summary Statistic	**Primary sensitivity**	**Reasoning**
Mean $\log x$: $\frac{1}{T}\sum \log x_t$	γ, δ	Approximates $\log x^* = \log(\gamma/\delta)$; sensitive to parameters governing prey equilibrium.
Mean $\log y$: $\frac{1}{T}\sum \log y_t$	α, β	Approximates $\log y^* = \log(\alpha/\beta)$; sensitive to parameters governing predator equilibrium.
Variance $\log x$	α, β	Captures amplitude of prey oscillations; larger α or smaller β produces wider swings.
Variance $\log y$	δ, γ	Captures amplitude of predator oscillations.
Cross-correlation $\log x$, $\log y$ at lag 0	All	Measures the phase relationship between predator and prey; a negative value is characteristic of predator–prey cycles.
Dominant oscillation frequency	α, γ	The linearised system oscillates at frequency $\omega \approx \sqrt{\alpha\gamma}$; sensitive to both growth and mortality rates.

Table 3: Summary statistics used for the Lotka-Volterra model, computed on log-transformed populations.

9.4 Validation Results

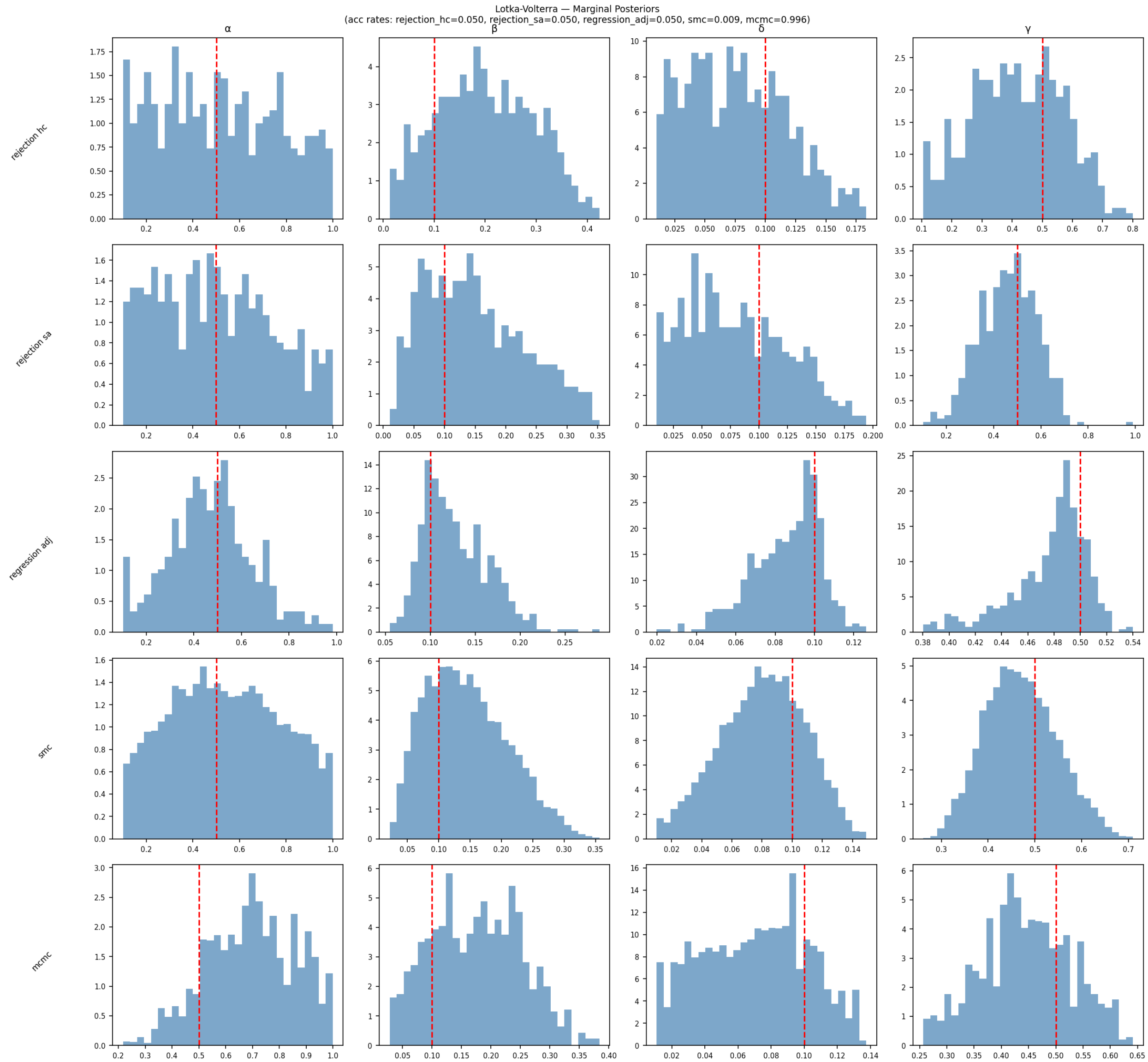

Figure 5: ABC posterior samples (orange) from each method with true parameters $\theta^* = (0.5, 0.1, 0.1, 0.5)$ marked with a red line. Rejection ABC (HC and SA) use $N = 10{,}000$ simulations at $q = 0.05$. SMC-ABC uses $M = 10{,}000$ particles over $T = 5$ stages. MCMC-ABC runs a chain of 10,000 steps at $\varepsilon = 1.0$. Regression adjustment is applied to the rejection ABC (HC) result. Synthetic likelihood uses $N = 2{,}000$ steps with $M = 200$ replicates per step.

Tables 4 and 5 summarises posterior means and 90% credible intervals for each method.

Method	$\hat{\alpha}$	$\hat{\beta}$	**90% CI** (α)	**90% CI** (β)
Rejection ABC (HC)	0.514	0.200	[0.130, 0.937]	[0.048, 0.353]
Rejection ABC (SA)	0.498	0.150	[0.141, 0.921]	[0.039, 0.300]
Regression Adj.	0.471	0.128	[0.174, 0.777]	[0.080, 0.192]
SMC-ABC	0.542	0.146	[0.169, 0.931]	[0.053, 0.262]
MCMC-ABC	0.695	0.176	[0.409, 0.942]	[0.058, 0.295]
Synthetic Likelihood	0.960	0.364	[0.959, 0.962]	[0.363, 0.366]
True θ^*	0.500	0.100	—	—

Table 4: Posterior means and 90% credible intervals for α and β for each method on the Lotka-Volterra model with $\alpha^* = 0.5, \beta^* = 0.1$.

Method	$\hat{\delta}$	$\hat{\gamma}$	**90% CI** (δ)	**90% CI** (γ)
Rejection ABC (HC)	0.077	0.417	[0.017, 0.148]	[0.167, 0.660]
Rejection ABC (SA)	0.080	0.462	[0.017, 0.157]	[0.262, 0.648]
Regression Adj.	0.086	0.477	[0.052, 0.109]	[0.414, 0.511]
SMC-ABC	0.080	0.470	[0.032, 0.125]	[0.349, 0.605]
MCMC-ABC	0.070	0.446	[0.019, 0.121]	[0.301, 0.584]
Synthetic Likelihood	0.461	0.223	[0.460, 0.462]	[0.223, 0.224]
True θ^*	0.100	0.500	—	—

Table 5: Posterior means and 90% credible intervals for δ and γ for each method on the Lotka-Volterra model with $\delta^* = 0.1, \gamma^* = 0.5$.

All rejection-based methods produce wide credible intervals spanning much of the prior support, reflecting the high-dimensional parameter space and the limited discriminating power of the summary statistics at $q = 0.05$. Regression adjustment noticeably tightens the intervals, particularly for β, δ, and γ, though at the cost of the calibration issues discussed below. MCMC-ABC produces an acceptance rate of 0.996, indicating the chain is accepting nearly every proposal and behaving as a prior random walk rather than concentrating on the posterior. This failure is discussed in detail in the MCMC-ABC subsection above. The synthetic likelihood results show extremely tight intervals far from the true parameter values ($\hat{\alpha} = 0.960$ versus $\alpha^* = 0.500$), a consequence of the chain never leaving its initialisation point due to the extreme concentration of the synthetic likelihood surface, as diagnosed in Section 6.

SBC Results

SBC was run over 200 trials with $L = 100$ posterior draws per trial, first on raw rejection ABC and then on the regression-adjusted result. Figures 6 and 7 show the rank histograms for each parameter.

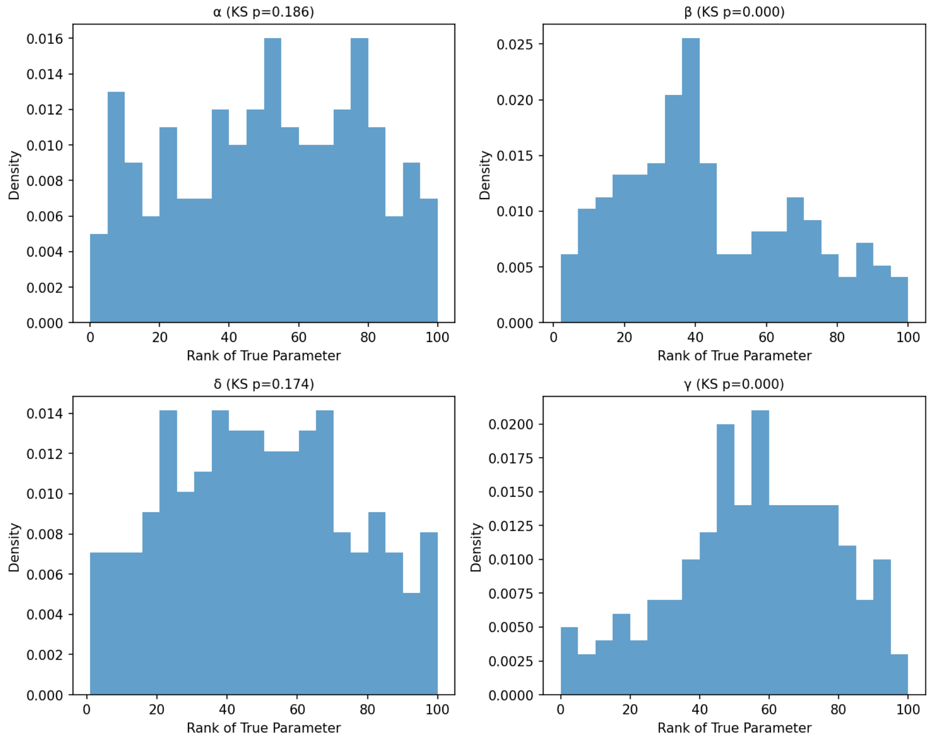

Figure 6: SBC rank histograms for α, β, δ and γ under rejection ABC at $q = 0.05$.

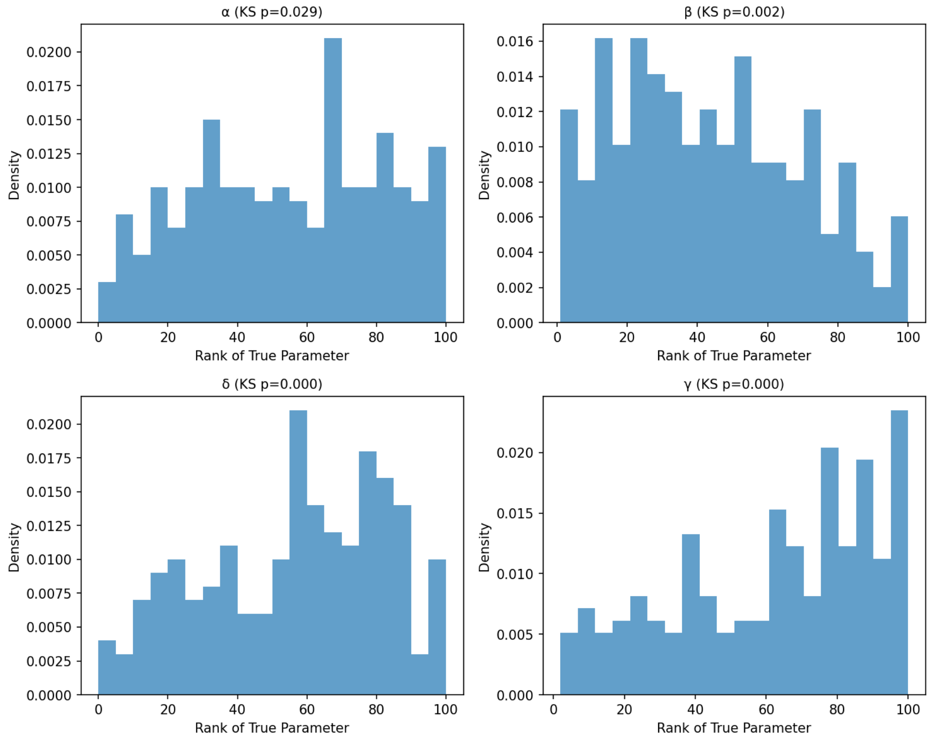

Figure 7: SBC rank histograms for α, β, δ and γ after regression adjustment.

Under raw rejection ABC, α and δ pass the KS test (p=0.186 and p=0.174 respectively) while β and γ fail strongly (p≈0). The rank histograms show a left-skew for β and a right-skew for γ, indicating the posterior systematically overestimates β and underestimates γ. This is consistent with summary statistic insufficiency: β and γ do not appear directly in the stationary point expressions $x^* = \gamma/\delta$ and $y^* = \alpha/\beta$, making the mean log-population statistics relatively insensitive to changes in these parameters individually.

Regression adjustment does not correct the miscalibration and in fact worsens it across all four parameters, with all KS p-values falling below 0.05 after adjustment. This confirms that the linear correction is misspecified for the Lotka-Volterra model, where the relationship between summary statistics and parameters is strongly non-linear. Unlike the MA(2) case where regression adjustment successfully corrected tolerance-induced bias in θ_2, the LV model violates the linearity assumption that underpins the correction.

STR Results

STR was run across a grid of 81 parameter values spanning the prior support. Figure 8 shows the credible intervals for each grid point.

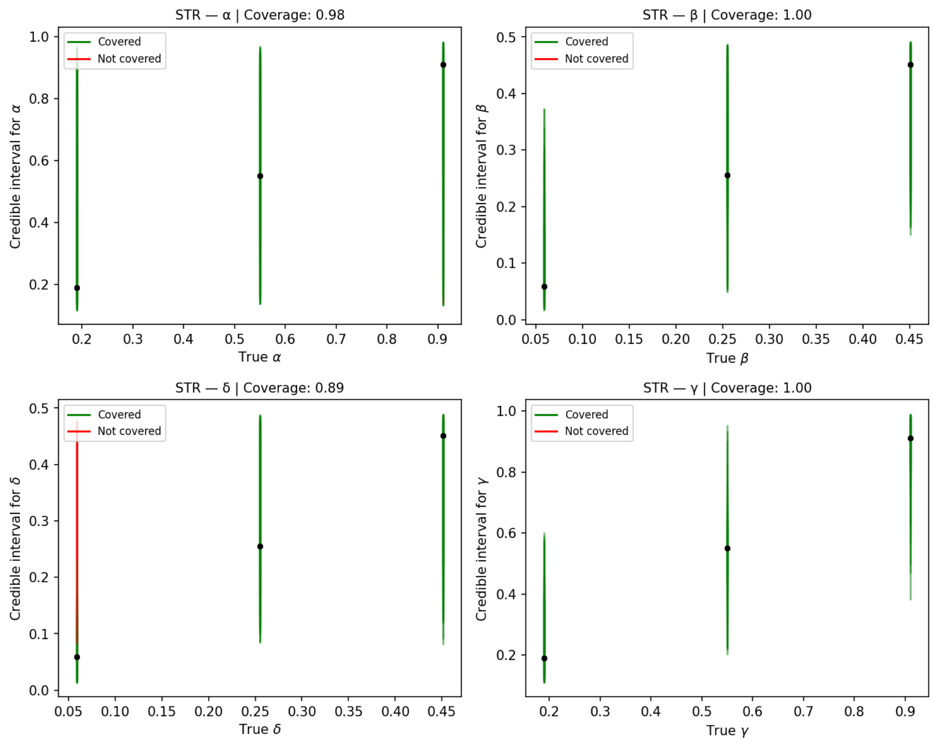

Figure 8: STR credible intervals across the parameter grid for α, β, δ and γ. Green intervals contain the true parameter; red intervals do not.

The empirical coverage is 0.975 for α, 1.00 for β, 0.889 for δ, and 1.00 for γ. The reduced coverage for δ is consistent with the SBC finding. The posterior underestimates δ, causing the credible interval to sit too high to contain the true value near the lower boundary of the prior support. The elevated coverage for α and the perfect coverage for β and γ reflect overdispersed posteriors, the wide credible intervals produced by rejection ABC at $q = 0.05$ are broad enough to contain the truth even when the posterior is not well-calibrated. This illustrates the complementarity of SBC and STR, SBC detects the systematic miscalibration that STR's broad intervals obscure.

PPC Results

PPC was run on the regression-adjusted posterior using sample mean and sample variance as test statistics. The posterior predictive p-values were 0.252 (mean) and 0.210 (variance), both well within the acceptable range. The model reproduces the mean and variance of the observed time series under the approximate posterior, providing no evidence of model misfit for these statistics. This result is not contradicted by the SBC failures, PPC checks whether the model can reproduce specific features of the observed data, while SBC checks global calibration across the prior. A miscalibrated posterior can still pass PPC if the test statistics are insensitive to the miscalibration direction.

10 Discussion and Decision Guide

The case studies in this report expose a consistent set of tensions that any practitioner of simulation-based inference must navigate: between simulation budget and posterior quality, between method complexity and tunability, between distributional assumptions and robustness. This section synthesises the comparative findings and translates them into actionable guidance.

10.1 Comparative Findings

Summary statistic sufficiency is the binding constraint

The most important lesson from both case studies is that no method can recover what the summary statistics do not contain. In the MA(2) model, the lag-1 and lag-2 autocorrelations are approximately sufficient for (θ_1, θ_2), and all methods produced reasonable posteriors. In the Lotka-Volterra model, the six summary statistics are insufficiently sensitive to β and γ individually, and the SBC rank histograms showed systematic miscalibration for these parameters regardless of which method was used. Tightening ε, switching samplers, or applying post-processing cannot recover information that the statistics do not capture. Summary statistic design is therefore the first and most important modelling decision, and SBC is the primary tool for diagnosing whether it has been done well.

SMC-ABC is the most consistently reliable method

Across both case studies, SMC-ABC produced the tightest and most concentrated posteriors among the rejection-based methods. Its adaptive tolerance schedule concentrates the particle population progressively, avoiding the inefficiency of a fixed ε while remaining robust to the dimensionality of the parameter space. Critically, it does not suffer from the degeneration problems that affected MCMC-ABC and synthetic likelihood in the four-parameter Lotka-Volterra setting. The trade-off is computation as SMC-ABC requires many more simulator calls than rejection ABC at the same number of accepted samples, and its tolerance schedule is sensitive to the choice of q and the number of stages T.

MCMC-ABC degenerates in higher dimensions with uniform priors

MCMC-ABC performed well on MA(2), producing tighter credible intervals than rejection ABC at comparable simulation budgets. On Lotka-Volterra it failed, with a uniform prior the Metropolis-Hastings acceptance ratio reduces to 1 inside the prior support, leaving the ε-gate as the only concentration mechanism. At the tolerances achievable within a reasonable simulation budget, the gate was too loose to concentrate the chain, and the samples were effectively prior draws. This is a known limitation of MCMC-ABC in higher-dimensional spaces [Marjoram et al., 2003]. Thus the method is best suited to low-dimensional problems where the posterior is concentrated relative to the prior and a tight ε is computationally feasible.

Synthetic likelihood requires careful numerical treatment

Synthetic likelihood worked well on MA(2) once the implementation was moved to log space, confirming the Gaussian assumption holds for the autocorrelation statistics. On Lotka-Volterra, the synthetic likelihood surface was extremely concentrated. The log-likelihood at the true parameters was approximately 20, while at randomly sampled prior draws it fell between -10^4 and -2×10^5. This extreme concentration caused the MCMC chain to become trapped at its initialisation point, producing a degenerate posterior. The concentration reflects accurate inference in principle. The Gaussian approximation assigns near-zero probability to any parameter vector inconsistent with the observed summaries, but makes exploration with a fixed Gaussian proposal practically impossible. Synthetic likelihood is most effective when the simulation budget can support large M (replicates per step), the summary statistics are approximately Gaussian, and the posterior is not so concentrated that standard proposals fail.

Regression adjustment corrects tolerance bias but not structural misspecification

In the MA(2) case study, regression adjustment successfully corrected the tolerance-induced bias in θ_2 detected by SBC, with the KS p-value increasing from ≈ 0 to 0.712 after adjustment. In the Lotka-Volterra case, regression adjustment worsened calibration across all four parameters. The difference is structural. MA(2) has an approximately linear relationship between its summary statistics and parameters near the posterior mode, making the linear correction appropriate. Lotka-Volterra's non-linear dynamics produce a non-linear summary-to-parameter mapping that a linear regression cannot capture. Regression adjustment should always be validated with SBC after application, its effectiveness cannot be assumed.

10.2 Decision Guide

The following guide is intended to help a practitioner choose an appropriate method for a new simulation-based inference problem.

1. **Start with a pilot.** Before choosing a method, run 500–1000 pilot simulations from the prior and examine the distribution of distances to s_{obs}. The 1%, 5%, and 50% quantiles tell

you what tolerances are achievable and how discriminating the summary statistics are. If the 1% quantile is large relative to the expected posterior width, the summary statistics may be insufficient.

2. **Choose a method based on dimensionality and simulation cost.**

   - *1-2 parameters, cheap simulator*: Any method is feasible. Rejection ABC with regression adjustment is the simplest starting point. MCMC-ABC is effective if the prior is non-uniform or informative.
   - *1-4 parameters, moderate simulator cost*: SMC-ABC is the recommended default. Its adaptive schedule produces tighter posteriors than rejection ABC without requiring manual epsilon tuning. Synthetic likelihood is viable if the summary statistics are approximately Gaussian and M can be set large enough to stabilise the covariance estimate.
   - *4+ parameters, expensive simulator*: Rejection ABC with regression adjustment is the most practical option, the simulation budget is fixed and the linear correction partially addresses tolerance bias. SMC-ABC remains viable but its per-accepted-sample cost is high. MCMC-ABC and synthetic likelihood are likely to degenerate and are not recommended without informative priors or strong summary statistics.
   - *Any dimensionality, very expensive simulator (>1 second per simulation)*: Neural methods such as Neural Posterior Estimation [Papamakarios and Murray, 2018] or Neural Likelihood Estimation should be considered. These methods amortise the simulation cost across observations by training a neural density estimator offline, eliminating the need for repeated simulation at inference time.

3. **Design summary statistics carefully.** Use domain knowledge to identify statistics that are sensitive to each parameter individually. Verify sufficiency using SBC before committing to a full inference run. Where possible, use semi-automatic ABC [Fearnhead and Prangle, 2011] to learn optimal linear combinations of candidate statistics from pilot simulations.

4. **Validate with all three diagnostics.**

   - *SBC* to check global calibration. Run before trusting any posterior. A failing SBC indicates either tolerance bias (correctable with regression adjustment) or summary statistic insufficiency (not correctable without better statistics).
   - *STR* to check instance-level recovery across the prior support. A sampler can be well-calibrated on average but fail near the boundaries of the prior, STR detects this.
   - *PPC* to check model adequacy. A passing PPC does not imply a well-calibrated posterior, but a failing PPC indicates the model cannot reproduce the observed data under any reasonable parameter values.

5. **Apply regression adjustment only if the summary-to-parameter relationship is approximately linear.** Validate the correction with SBC after applying it. If SBC worsens after adjustment, the linearity assumption is violated and the correction should not be used.

10.3 Limitations and Future Directions

The most significant limitation of the methods covered in this report is their dependence on hand-crafted or semi-automatically selected summary statistics. As demonstrated in the Lotka-Volterra case study, insufficient statistics produce miscalibrated posteriors that no sampler can correct. Neural simulation-based inference methods address this directly by learning sufficient statistics end-to-end from simulations, without requiring domain expertise or manual selection. Neural Posterior Estimation, Neural Likelihood Estimation, and Neural Ratio Estimation represent the current frontier of simulation-based inference [Cranmer et al., 2020, Papamakarios and Murray, 2018] and would be natural extensions of the `abclib` library.

A secondary limitation is the Euler-Maruyama discretisation used for the Lotka-Volterra simulator. The fixed step size $\Delta t = 0.1$ introduces discretisation error that is not accounted for in the inference, potentially contributing to the observed miscalibration. A more accurate simulator using an adaptive ODE solver, or a stochastic kinetic model using the Gillespie algorithm, would reduce this source of error at increased computational cost.

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